Tasbeeh Hasan

Computer Engineering Student • AI, Quantum Computing & Research

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PROFESSIONAL SUMMARY

Research-driven computer engineering student (Barnard University remote scholar program) with advanced training in quantum computing, machine learning, and applied physics. Ranked 1st globally in the 2026 Purple Comet Mathematical Challenge (team), 22nd worldwide at IPhR 2025, and 3rd nationally at EOCS 2025. Built AI projects including GlycoPal, a retrieval-augmented generation (RAG) system on insulin resistance, and a multi-agent reinforcement learning model for personalized cancer treatment (Top 5, AppX 25). Team lead, tutor, and mentor, with 40+ participants mentored. Seeking research, internship, and technical opportunities in AI, quantum computing, and engineering.

EDUCATION

Computer Engineering, Remote Scholar Program Ongoing
Barnard University

Digital Egypt Cubs Initiative (DECI) Scholarship: AI & Data Science, Level 4 2022 to present
Ministry of Communications and Information Technology (MCIT), Egypt

- Fully funded government scholarship strengthened leadership and entrepreneurship skills through the annual summer camp.

RESEARCH & PROJECTS

GlycoPal: RAG System for Insulin Resistance Creativa AI Hackathon

- Developed a complete retrieval-augmented generation (RAG) project focused on insulin resistance.

Vertical-Axis Wind Turbine: National Finalist, i-club Competition 2024

- Designed a novel turbine in SOLIDWORKS, simulated it in ANSYS, and validated it with mathematical and physics-based analysis.
- Secured Innovation Support Fund (ISF) funding for prototype development; co-authoring a research paper and preparing a patent application with a teammate.

Personalized Cancer Treatment Model: Top 5, Bioinformatics Track, IEEE Zewail City AppX 25 Feb 2025

- Built a multi-agent reinforcement learning model (MADDPG) integrating multi-omics data (team Genomind).

"Evolutionary Perspective on Brain Efficiency": 2nd Place, STEM Quest Research Competition Jan 2025

- Authored an internationally reviewed LaTeX paper linking evolutionary biology, neuroscience, AI, and robotics.

"The Role of Magnetism in Earthquake Dynamics and Prediction" Feb 2025

- Wrote an 18-page independent analysis of global seismic-magnetic correlations, identifying magnetic field anomalies that precede seismic events.

Exoplanet Detection: Paradox Team, PACS e-Lab & AAVSO Aug 2025

- Analyzed light curves and telescope data and validated candidate exoplanets as part of a Pan-Africa research collaboration.

Spherical UAV Drone: Robotics Championship 2025

- Designed and built a spherical UAV drone for a robotics championship.

LEADERSHIP & EXPERIENCE

Team Lead, Innov Egypt Program Sep 2025

- Led a nine-member team from ideation to a final pitch that earned an A+ evaluation.

Team Member & Astronomy Tutor, Paradox 2025

- Organized three cohorts of International Astronomical Search Collaboration (IASC) asteroid search campaigns, mentoring 40+ participants in data analysis and asteroid discovery.
- Designed and delivered astronomy lessons for students in the Space Scholar Program.
- Represented the team at the African Astronomical Society (AfAS) online conference and Pan-Africa outreach events.

Web Developer (Self-Directed) 2025 to present

- Build interactive science and education websites from scratch in HTML, CSS, and JavaScript, without frameworks, including a personal portfolio.

Volunteer, Resala Organization 2025

Campus Ambassador, Winter of Blockchain, GirlScript Foundation 2024

ACHIEVEMENTS & AWARDS

- **1st Place worldwide, Purple Comet Mathematical Challenge** (international team competition, 2026)
- **Silver Honor, International Physics Realm (IPhR) 2025**, global rank 22
- **3rd Place, Egyptian Olympiad in Computational Science (EOCS)**, Final Round (Aug 2025)
- **Bronze Honor, International Astronomy and Astrophysics Competition (IAAC)**, qualified for the Final Round (Sep 2025); result confirmed in 2026
- **National Finalist, Intel/Regeneron ISEF Egypt** (2025)
- **3rd Place, Creativa Sustainability Hackathon** (2025)
- **National Finalist, i-club Competition** (2024)
- **Selected Technology Talent, Ebhar Masr Program** (Banque Credit Agricole Egypt Development Foundation, Jul 2025), one of six technology talents nationwide and 21 across all fields
- **Most Inspirational & Best Presentation Awards**, NASA Space Apps Challenge, Benha (Oct 2022)

CERTIFICATIONS & TRAINING

- **ADEQUATE Quantum Computing Course** (QURECA & QWorld, EIT Deep Tech Talent Initiative, Aug 2025): 50+ hours covering Shor's, Grover's, VQE, QAOA, and QPE algorithms, error correction, and quantum ML; ran circuits on real quantum hardware.
- **Quantum Summer School and Air Quality Prediction Competition** (Nile University & ASRT, Aug 2025): quantum computing with Qiskit and PennyLane applied to satellite-based environmental monitoring.
- **Womanium & WISER Online Quantum Program** (2025)
- **IEEE International Conference on Quantum Computing and Engineering (QCE)**, Albuquerque, USA (2025); **IEEE Nuclear Energy Conference** (2025)
- **Creativity and Innovation in Education Scholarship** (UNESCO Grant Scheme, Educuality, Aug 2025): 20 credit hours; team project on improving government school teachers' technology skills
- **Shape the Future Challenge** (Moonshot Pirates, Jul 2025): designed solutions for SDG 7 and SDG 11
- **Summer Programs (2025):** Arab Physical Society High Energy Physics (Zewail City); Physics (Tanta University); Physics (Cairo University)
- **Programs:** The Junior Academy, New York Academy of Sciences (Fall 2025); YSJ Annual Junior Research Program (2025); October Math Circle Calculus Camp (2025); Girls With Impact Entrepreneurship Program (2025)

SKILLS

Programming	Python (Qiskit, PennyLane, NumPy, Pandas, Matplotlib, TensorFlow), MATLAB, LaTeX, HTML, CSS, JavaScript
AI & Quantum	Reinforcement learning, retrieval-augmented generation, model development, quantum gates, error correction, quantum neural networks
Engineering Tools	SOLIDWORKS, ANSYS, AutoCAD
Data & Research	Scientific visualization, experimental design, literature review, hypothesis formulation, proof writing
Communication	LaTeX papers and posters, oral and written presentations at international events, mentoring, public speaking
Working Style	Leadership, teamwork, adaptability, problem-solving, growth mindset

LANGUAGES

Arabic (Native) • English (Advanced working proficiency) • German (Elementary)